

QUESTION 1

- (a) Find all real solutions to $x + 6 > x^2$.
- (b) Differentiate $f(x) = \cos^{-1}(\sin x)$.
- (c) Find k if $x^{k+3} = e^{7 \ln x}$, where $x > 0$.
- (d) Prove that $\frac{1 + \cos 2\theta}{1 - \cos 2\theta} = \cot^2 \theta$.
- (e) A firm manufacturing fuses finds that 2 % of them are defective.
From a random sample of 8 fuses, find the probability that the sample contains 3 defective fuses. Give your answer correct to one significant figure.

QUESTION 2 (START A NEW PAGE)

- (a) Use the substitution $u = 1 + 3x^3$ to evaluate $\int_0^1 x^2 \sqrt{1+3x^3} \, dx$.
- (b) Integrate $\int \frac{x^2 + 1}{x^2 + 4} \, dx$.
- (c) Prove by Mathematical Induction that :

$$1 + 4 + 16 + \dots + 4^n = \frac{1}{3} [4^{n+1} - 1] \text{ for } n = 0, 1, 2, \dots$$

J.R.A.H.S. TRIAL HSC EXAMINATION 3 / 4 UNIT MATHEMATICS 1993

QUESTION 3 (START A NEW PAGE)

- (a) A particle moving in a straight line x centimetres from the origin O , after t seconds, is given by $x(t) = 3 - 5 \cos 2t$.
- (i) Show that its acceleration is given by : $\ddot{x} = -4(x - 3)$.
- (ii) Find its period of motion.
- (b) There are 5 girls and 6 boys in a group.
- (i) How many ways could they be arranged in a line such that each girl stands between 2 boys?
- (ii) How many arrangements are possible if two boys A and B stand at each end of a line?
- (c) (i) Express $\cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$ for $R > 0$ and α acute.
- (ii) Hence, or otherwise, find all solutions to $\cos x - \sqrt{3} \sin x = 2$.

QUESTION 4 (START A NEW PAGE)

- (a) (i) Neatly sketch the graph of $y = \sin^{-1} x$ and state its domain and range.
- (ii) By considering the graph in (i) or otherwise, find the exact value of:

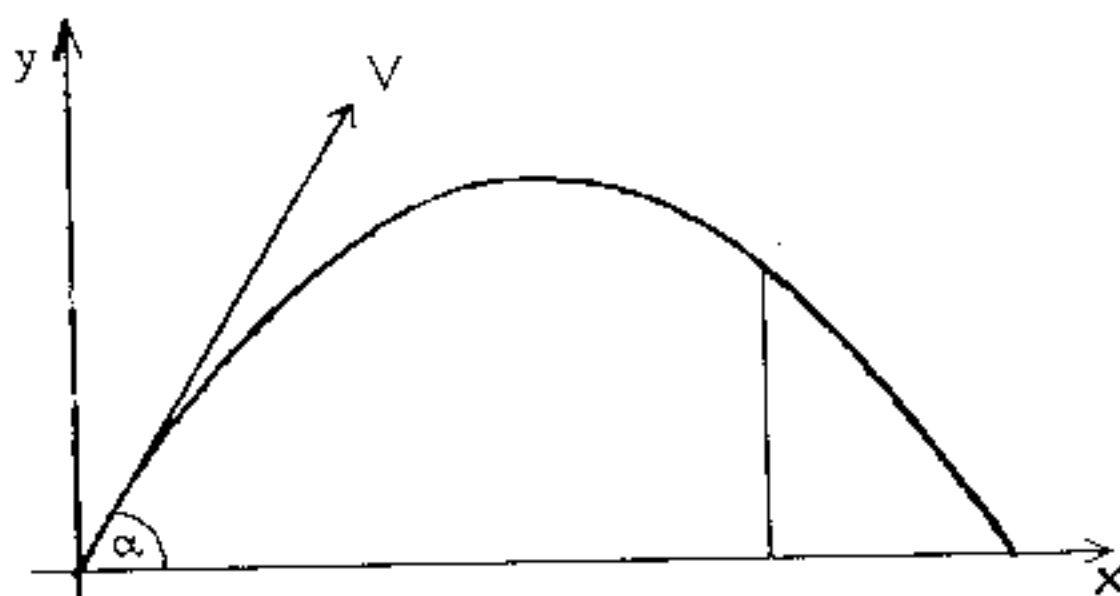
$$\frac{1}{2} \int_0^{\frac{1}{2}} \sin^{-1} x \, dx .$$

- (b) (i) Show that $\frac{d}{dx} \tan^3 x = 3 \sec^4 x - 3 \sec^2 x$.

- (ii) Using (i) or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x \, dx$.

QUESTION 5 (START A NEW PAGE)

- (a) A particle is projected from a fixed point O on a horizontal plane at an angle of elevation α with a speed of V m/s. After a time t , the horizontal and vertical components of its velocity are: $\dot{x} = V \cos \alpha$ $\dot{y} = V \sin \alpha - gt$



- (i) Show that the position $P(x,y)$ of the particle at any time as it moves along its path is given by:

$$y = x \tan \alpha - \frac{gx^2}{2V^2} (1 + \tan^2 \alpha)$$

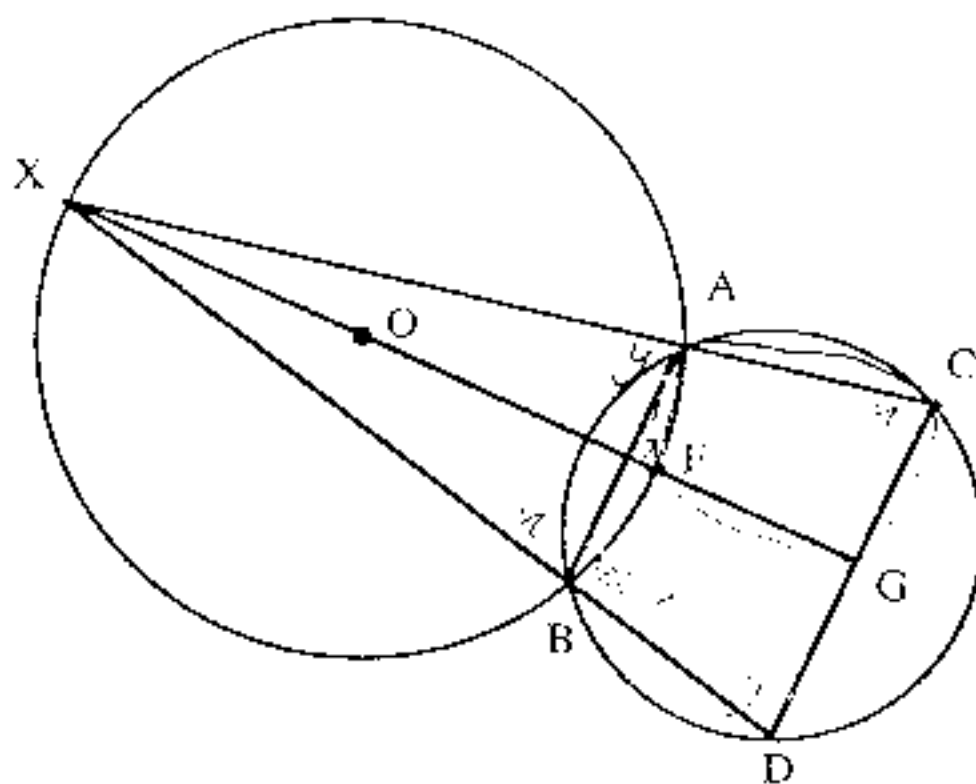
- (ii) If the particle is projected from the origin at an angle of 30° , find the speed required for it to just clear a vertical wall 4 metres high and 12 metres away from the origin. (Take $g = 10 \text{ m/s}^2$)
Give your answer correct to two decimal places.

- (b) (i) Write down the term T_{r+1} of the expansion $(x + \frac{1}{x^2})^{12}$ for $0 \leq r \leq 12$.

- (ii) Evaluate the term independent of x in the expansion of $(x + \frac{1}{x^2})^{12}$.

QUESTION 6 (START A NEW PAGE)

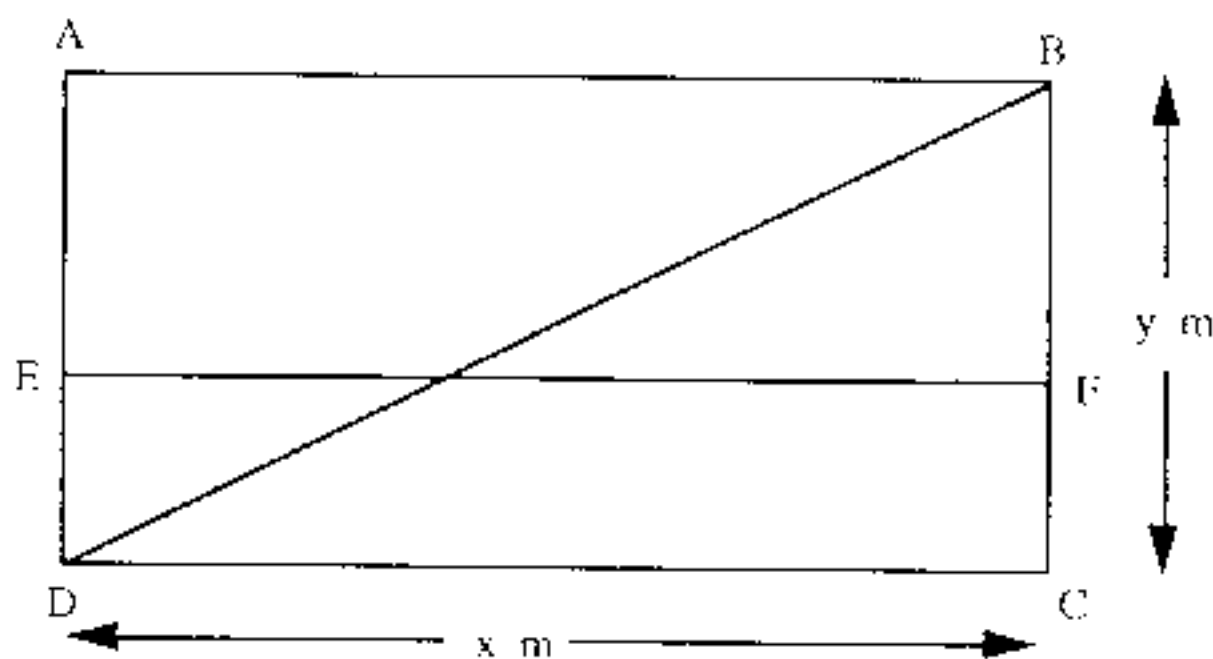
- (a) Prove that $\frac{d^2x}{dt^2} = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$.
- (b) The acceleration of a particle x metres from origin O at time t seconds is given by $\ddot{x} = -\frac{1}{2}e^{-x}$. If its velocity v , is 1 metre / second when $x = 0$, find its velocity when $x = 4$.
- (c) Two circles cut at A and B . X is on the circle with centre O and XA, XB cut the other circle at C, D respectively. XO extended cuts the circle XAB at F and chord CD at G .



- (i) Neatly copy the diagram onto your answer sheet.
- (ii) Prove that $ACGF$ is a cyclic quadrilateral.
- (iii) Prove that XG is perpendicular to CD .

QUESTION 7 (START A NEW PAGE)

- (a) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$ and $p + q = 2$.
- Find the angle which PQ makes with the x axis.
 - Find the equation of the locus of M , the mid-point of PQ as it moves about the parabola.
 - Clearly state any restrictions on the locus of M .
- (b) A rectangle $ABCD$ with sides of length x metres and y metres has an area of 9 m^2 . Two metal construction strips, one a diagonal BD and the other EF parallel to sides AB and CD are required to strengthen the rectangle.



- Show that the total length L , of BOTH strips is :
$$L = x + \frac{\sqrt{x^4 + 81}}{x} \text{ metres.}$$
- Find the dimensions of the rectangle which will minimise the total length L of the strips.

THIS IS THE END OF THE PAPER

Q1 (a) $-2 < x < 3$ 4a (i) $-1 \leq x \leq 1$
 (ii) $-1, \cos x > 0$ R: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
 (iii) $\cos x < 0$
 Undefined $\cos x = 0$ (ii) $\frac{\pi}{12} + \frac{\sqrt{3}}{2} = 1$

(c) $\frac{1}{4} = 4$ 4a (i) $-$
 (ii) $4/3$
 (d) $-$
 (e) 0.0004 5a (i) $-$
 (ii) 18.11 m/s

Q2 (a) $\frac{12}{27}$ 6 (i) $\left(\frac{12}{27}\right)^{35+2}$
 (ii) $T_1 = 495$
 (b) $x = \frac{3}{2} \tan^{-1} \frac{x}{2} + C$

(c) $-$ 6 (a) $-$
 (b) $V = \pm \frac{1}{2} \ln x$

Q3 (a) $-$ (c) $-$
 (ii) $T = \pi$

4a (i) 86.40° 7a (i) 1
 (ii) $72.5-76^\circ$ (ii) $x = 2a$
 (iii) $y \geq a$

(c) (i) $R = 2, \alpha = \pi/3$ (b) (i) $x = 3^{3/4}, y = 3^{3/4}$
 (ii) $x = 2n\pi - \pi/3$